



Prevalence of polymyxin resistance through the food chain, the global crisis

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Abstract

Antimicrobial resistance is one of the vital challenges facing global health today. Multi-drug resistant (MDR) infections are often treated with the narrow-spectrum drugs, colistin (polymyxin E) or polymyxin B, which are last-resort antibiotics for human therapeutics that are effective against Gram-negative bacteria. Unfortunately, resistance to these polymyxins has occurred because of selective pressure caused by the inappropriate use of those antibiotics, especially in farming. The mechanisms of resistance to polymyxins are mediated through intrinsic, mutational, or genetic alteration in chromosomal genes. The mechanism includes the regulatory network controlling chemical modifications of lipid A moiety of lipopolysaccharide, reducing the negative charge of lipid A and its affinity for polymyxins. Additionally, the unique mobile colistin/polymyxin B resistance (*mcr*) gene reported in Enterobacteriales is responsible for the horizontal dissemination of resistance to polymyxins *via* the food chain. There is now an urgent need to increase surveillance for detecting resistance to polymyxins. Therefore, this review presents an overview of presently available scientific literature on the mechanism of resistance to polymyxins, with their associated gene variants, evaluation methods, resistance transmission through the food chain *via* food bacteria, and related risk factors. We further focus on the significant implications of polymyxins usage in India and future views for food safety to preserve polymyxin activity.

Introduction

Antimicrobial resistance has become a global health care emergency with the presence of numerous multi-drug resistant (MDR) pathogens [1]. Reports suggest that antibiotic resistance could be one of the prime reasons for 10 million annual deaths from infections by 2050 [2, 3]. The increase in antimicrobial resistance is a phenomenon interconnected with the overuse or abuse of antibiotics that promote the selection of resistant bacteria [4]. The scarcity of potent, newly approved antibiotics in the last decade has led to the increasing usage of polymyxins to treat multidrug-resistant Gram-negative bacterial infections

[5, 6]. Polymyxins, a class of cationic cyclic polypeptide antimicrobials, discovered during the late 1940s, acts as a “last-resort” antibiotic choice in treating infections caused by carbapenem-resistant Gram-negative pathogens [1, 7, 8]. Colistin and polymyxin B fall under the polymyxin antibiotic group, produced by *Bacillus* sp., and organisms resistant to one are completely cross-resistant to the other polymyxin [8]. Their mechanisms of action are the same, but structurally, they differ only in one amino acid, i.e., D-leucine in polymyxin E (colistin) is replaced by D-phenylalanine in polymyxin B. Polymyxin B is available for parenteral use as the sulfate salt and preferred for routine systemic use whereas, colistin is administered as the sodium salt of colistin methanesulphonate and recommended for use for the treatment of urinary tract infections [9].

For the purpose of this review, the term polymyxin is used to refer to both or either polymyxin B and/or colistin. Polymyxins are also popular drugs in farming where they are used as a growth promoter and also to prevent and treat infections in animals [10]. The rapid spread of colistin/ polymyxin-resistant genes through plasmid-mediated determinants has been identified. This phenomenon has led to the prohibition of the use of polymyxins in animal husbandry as growth

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promoters and for prophylactic use, in order to preserve the drug's effectiveness for future human use [11–13].

Bacteria gain resistance to polymyxins by employing various strategies. These strategies include alteration of their lipid A, mainly by the modification of 4-amino-4-deoxy-L-arabinose (L-Ara4N) and phosphoethanolamine (PEtN), by hydroxylation, and diacylation, the utilization of an efflux pump, capsule formation and the loss of lipopolysaccharide, etc. [14, 15]. The movement of resistance genes from farm to fork through the food chain to humans has led us to focus on this topic. Although many reports have described the mechanisms of polymyxin resistance [11, 16], in this manuscript, we provide a comprehensive and updated review of various mechanisms of polymyxin resistance. This review also describes the predominance of the plasmid-borne mobile colistin resistance (*mcr*) genes and their allelic variants. Additionally, we discuss the One Health concept to understand the judicious use of polymyxins in animal husbandry, and in humans. In addition, this review article also provides detailed information on the misuse of polymyxins in India and the regulatory measures that have been undertaken.

Usage of polymyxins as therapeutic agents in human and veterinary medicine

Colistin was discovered in 1947, and it was first commercialized in Japan and Europe during the 1950s [7]. It was subsequently authorized by the U.S. Food and Drug Administration in 1959, replacing it with the inactive prodrug, colistimethate sodium (CMS) of colistin, for parenteral use [7]. Polymyxins have a narrow range of antimicrobial action, specifically against multidrug-resistant (MDR) Gram-negative Enterobacteriales and *Pseudomonas aeruginosa* [4, 11, 17]. During the 1970s, colistin use was reconsidered because of its toxicity, especially nephrotoxicity [18], and its use was restricted to topical and ophthalmic therapies for over 20 years. Nebulized or systemic polymyxin was primarily used in cystic fibrosis patients. Later, the growing incidence of multidrug-resistant (MDR) Gram-negative bacteria [19] has led to the increased use of the polymyxin antibiotics [7, 11, 18]. Polymyxin B was approved by the Food and Drug Administration for clinical use in the United States in 1964. The polymyxins are now considered as last-line defence antibiotics against MDR organisms, predominantly carbapenemase-producing bacteria.

The clinical use of colistin and polymyxin B are different and are not interchangeable. Colistimethate sodium (CMS), an inactive prodrug of colistin, is converted to active drug in the kidneys. Colistin is mostly used to treat urinary tract infections as it is in the pro-drug form and is not bioavailable as active drug in the bloodstream. In contrast, polymyxin B is administered directly in its antibacterial form, is active in the blood and is excreted by the liver and is not

available in the kidneys/urine. Therefore, polymyxin B is used for bloodstream and systemic infections and colistin primarily for urinary tract infections [9, 18].

Polymyxins have also been used for decades in veterinary medicine to prevent and treat infections. Polymyxins were prescribed in veterinary medicine for oral use to act in the gastrointestinal tract to treat MDR enterobacterial infections and gastrointestinal disorders caused by *E. coli* in poultry and pig farming [20]. Polymyxins are used as animal growth promoters in farming worldwide. The excessive use of these antibiotics has led to the development of emergence of resistance in farm animals. Polymyxins are used as animal growth promoters in farming worldwide. The usage of polymyxins differs from country to country. The United States government has banned the use of polymyxin/colistin in animal husbandry due to the need to preserve it for human use [21]. China has the highest consumption of polymyxins in agriculture, however, in recent years China has also banned the use of polymyxins use as a growth promoter in animal husbandry [22, 23]. Polymyxin usage is also high in Germany, Estonia, Italy, and Portugal but more recently these countries have also regulated the usage of polymyxins in animals [24, 25].

Resistance Mechanisms for polymyxin in Gram-negative bacteria

Polymyxin resistance in multidrug-resistant bacteria is due to mutation and adaptation resulting from selection pressure [4]. Adaptive polymyxin resistance occurs after in vitro exposure [11]. However, mechanisms of resistance linked to chromosomal mutations cannot be transferred through horizontal gene transfer [15, 26, 27]. Resistance to polymyxins is mostly the result of modifications in the lipopolysaccharide layer of the bacterial cell wall and its reduced negative charges leading to resistance [28, 29]. Other mechanisms of polymyxin resistance are the overproduction of efflux pump system in the bacterial cell, overproduction of the capsule polysaccharide, and production of a polymyxin degrading enzyme by some organisms, like *Paenibacillus polymyxa* [18, 30, 31]. Polymyxin resistance through plasmid-mediated *mcr* genes in Gram-negative bacteria depends on intrinsic or mutational and adaptive mechanisms through horizontally acquired resistance [12, 32].

Genes encoding bacterial LPS-modifying enzymes: The operon (*pmrCAB* and *pmrHFIJKLM*), The gene (*pmrC*, and *pmrE*), and the two-component system (*PmrAB* and *PhoPQ*)

The polymyxin resistance pathway involves two operons. The *pmrCAB* encodes three proteins such as phosphoethanolamine (pEtN) phosphotransferase PmrC, PmrA,

which acts as a response regulator (known as BasR), and PmrB, which acts as a sensor kinase protein (known as BasS). The *pmrHFIJKLM* operon (also known as the *arnBCADTEF* or *pbgPE* operon) encodes seven other proteins [18, 33] (Table 1). The phosphoethanolamine phosphotransferase PmrC protein adds a pEtN cationic group to the bacterial LPS [33]. The *pmrE* and *pmrC* genes are responsible for the cationic L-aminoarabinose (L-Ara4N) group synthesis and its addition to bacterial lipid A via the *pmrHFIJKLM* operon [4]. Moreover, various environmental stimuli such as increased concentrations of ferric (Fe^{3+}) iron, aluminum (Al^{3+}) and decreased pH (e.g., pH 5.5), simultaneously induce PmrB protein. PmrAB and PhoPQ, the two-component systems, then prevent bacterial phagocytosis by macrophages [33]. Protein PmrB has tyrosine kinase activity that activates PmrA protein through phosphorylation. Activated protein PmrA regulates the transcription of the *pmrCAB* operon, the *pmrHFIJKLM* operon, and the *pmrE* gene which are involved in bacterial LPS modification (pEtN and L-Ara4N addition) [4, 12, 34] (Table 1). *PmrA* and *pmrB* gene mutation causes acquired polymyxin resistance in Enterobacteriales through the expression of the PmrAB two-component system, directing the upregulation of the *pmrCAB*, *pmrHFIJKLM* operon, and the *pmrE* gene, and promoting the production of cationic pEtN and L-Ara4N and their addition to LPS [15, 35–37] (Fig. 1). The operon *arnBCADTEF* is responsible for intrinsic polymyxin resistance in *S. marcescens*, resulting from mutations in *arnB* and *arnC* genes [15]. A two-component system, *phoPQ* operon, encodes PhoP, a regulator protein, and PhoQ, the sensor protein kinase. Decreased levels of magnesium (Mg^{2+}), calcium (Ca^{2+}) low pH (e.g., pH 5.5) promote the activation of sensor protein kinase PhoQ [18, 33, 34]. *PhoPQ* operon stimulates the expression of genes that support the magnesium transport system, activates enzymes that modify the bacterial LPS, and activates enzymes that downregulate the cell stress response caused by various acidic pH, all of which aids to resist cationic polymyxins [38]. Therefore, the operon *PhoPQ* two-component system helps bacterial survival under adverse situations of low magnesium or acidic pH or exposure to the cationic antimicrobial peptides including the polymyxins. PhoQ is a protein having tyrosine kinase activity which activates PhoP by phosphorylation. PhoP, then, allows the transcription of the *pmrHFIJKLM* operon, which contributes by adding L-Ara4N lipopolysaccharide [38] (Fig. 1). Either directly or indirectly PhoP can activate the PmrA protein, via the PmrD connector protein, that adds pEtN to LPS [39, 40]. Acquired resistance to polymyxin depends on several mutations in the *phoP* and *phoQ* genes that are responsible for constitutive activation of the *PhoPQ* two-component system [38]. The *PhoPQ* two-

component system helps in the upregulation of the *pmrHFIJKLM* operon and supports the synthesis of L-Ara4N and its transfer to lipid A of lipopolysaccharides membrane (Fig. 1). Other modifications linked to chromosomal polymyxin resistance have been elaborated, such as reducing the amount of acyl moieties via *lpxR*-like deacylation and hydroxylation of lipid A moiety [41]. Besides the acylation of lipid A, these modifications can change the permeability barrier properties of the outer membrane [42].

Polymyxin resistance mediated by *mgrB* gene

MgrB acts as a transmembrane protein consisting of forty-seven amino acids. The *mgrB* gene is activated via the activation of the PhoP regulatory protein [43]. The MgrB protein minimizes the expression of the PhoQ-encoding gene, leading to downregulation of the *PhoPQ* two-component system (Fig. 1) [43]. Inactivation of the *mgrB* gene, in turn, upregulates the expression of the *phoPQ* operon, thereby activating the *pmrHFIJKLM* operon, leading to the induction of L-Ara4N responsible for gaining polymyxin resistance. Other mutations, like deletions or insertions of some nucleotide sequences or mutation in the locus of *mgrB* gene cause extrinsic resistance to polymyxin [15, 18, 44, 45] (Table 1).

Resistance through the *crrAB* operon

The polymyxin resistance regulatory *crrAB* operon consists of two proteins, CrrA, which acts as a regulatory protein; another is CrrB, which acts as a sensor protein kinase. To gain polymyxin resistance, inactivation of the *crrB* gene positively regulates the expression of the *pmrAB* two-component system, which helps in stimulation of the *pmrHFIJKLM* operon and the associated *pmrC* and *pmrE* genes, subsequently directing the synthesis of L-Ara4N and pEtN, that leads to gain polymyxin resistance [46] (Fig. 1). Moreover, inactivation of CrrB, sensor protein kinase also alters the lipid A moiety via stimulation of a glycosyltransferase-like protein leading to resistance to polymyxin [46] (Table 1).

Resistance via the presence of a capsule

The bacterial capsular layer containing polysaccharides protects the organisms against polymyxins [47] (Table 1). Several bacteria like *Klebsiella pneumoniae* possess capsular polysaccharides on their surface [48]. The capsular layer on the cell surface enhances polymyxin resistance by acting as a barrier to the penetration of the antibiotic [47]. The impact of the capsular biosynthesis gene on decreasing the interaction of polymyxin with the lipid

Table 1 Colistin resistant mechanisms, associated variants and organisms

Resistance mechanisms	Genes/Operons	Reported Organisms	Cell's modification	Report from	References
Modification of the lipid A with aminoarabinose & phosphoethanolamine (L-4AraN and pEIN)	<i>arnBCADTEF</i> operon and <i>pmrE</i> , <i>pmrC</i> gene	<i>S. enterica</i> , <i>E. coli</i> , <i>Serratia marcescens</i> , <i>P. mirabilis</i> , <i>P. aeruginosa</i> , <i>Burkholderia</i>	Constitutive overexpression of mutated TCS causes permanent alteration of lipid A via L-Ara4N and pEIN. LPS increases their surface charge in the presence of high Zn^{2+} that lower the binding ability of polymyxin, gaining the resistance against the antibiotic	Canada, Switzerland	[15, 18, 54]
Activation of LPS- <i>via</i> alteration of <i>arnBCADTEF</i> operon	<i>pmrA</i> / <i>pmrB</i>	<i>K. pneumoniae</i> , <i>Enterobacter aerogens</i> , <i>E. coli</i> , <i>S. enterica</i> serovar <i>Typhimurium</i> , <i>A. baumannii</i>	Overexpression of <i>pmrCAB</i> and <i>pmrHFIJCLKM</i> operons <i>via</i> Substitution mutation in <i>PmrB</i> protein is responsible for resistance to colistin. Also, modification in the <i>pmrA</i> gene alone results in colistin resistance	France, Switzerland, Canada	[4, 36, 127]
Alteration in lipid A by modification of the <i>pmrHFIJCLKM</i> operon/activation of <i>pmrAB</i> <i>via</i> <i>pmrD</i>	<i>phoP</i> / <i>phoQ</i>	<i>P. aeruginosa</i> , <i>E. coli</i> , <i>K. pneumoniae</i> , <i>Enterobacter cloacae</i> , <i>Salmonella</i> sp	<i>PhoPQ</i> , activates <i>PmrAB</i> , <i>via</i> stimulating <i>pmrD</i> . The <i>pmr</i> operon is liable for aminoarabinose modification of LPS which protects from antimicrobial mediated disruption and killing. <i>PhoP</i> , response regulator, can stimulate L-Ara4N biosynthesis and addition of <i>phoPQ</i> to lipid A. Mutation in <i>PhoP</i> activates <i>pmrHFIJCLKM</i> , able to resist polymyxin <i>via</i> the production of L-aminoarabinose.	Canada, USA, Switzerland,	[126–129]
overexpression or alteration of <i>phoP</i> / <i>phoQ</i> regulatory system and stimulation of <i>pmrHFIJCLKM</i> operon	<i>mgrB</i>	<i>K. pneumoniae</i>	Alteration in the <i>mgrB</i> gene, upregulate the expression of <i>phoP</i> , <i>phoQ</i> , and <i>pmrK</i> finally activation of <i>pmrHFIJCLKM</i> operon, transcription, leads to resistance to polymyxins by modification in the lipopolysaccharide target.	Italy, Spain	[45]
Alteration of lipid A by amino acid substitutions in <i>CrrB</i>	<i>crrB</i> , <i>crrC</i>	<i>K. pneumoniae</i>	Amino acids change in <i>CrrB</i> encourage <i>CrrC</i> expression, subsequently causing upregulated <i>pmrHFIJCLKM</i> operon and <i>pmrC</i> expression that yielding increased colistin resistance.	Taiwan	[130]
An increased concentration of RamA alters the Lipid A biosynthesis, thus reduce bacterial adhesion and uptake into macrophages.	<i>ramA</i>	<i>K. pneumoniae</i>	<i>RamA</i> binds directly and promotes the <i>lpxC</i> , <i>lpxL-2</i> , and <i>lpxO</i> genes related to lipid A biosynthesis. This alteration increases the resistance to cationic antimicrobial peptide, polymyxin B and colistin.	United Kingdom	[57]
Alteration of lipopolysaccharides at the outer KDO residue by <i>eptB</i> gene that encodes enzyme phosphoethanolamine transferase	<i>eptB</i> , <i>phoP</i> / <i>phoQ</i> and <i>mgrR</i>	<i>E. coli</i>	After activation of <i>EptB</i> , it alters the 3-deoxy-D-manno-octulosonic acid (Kdo) residue of LPS where Phosphatidylethanolamine acts as a donor substrate. The binding reduces the net negative charge of molecules and decreases the electrostatic repulsion involving neighboring lipopolysaccharide molecules that helps to resist the polymyxin B activity.	USA	[17, 131]

Table 1 (continued)

Resistance mechanisms	Genes/Operons	Reported Organisms	Cell's modification	Report from	References
Mutation in <i>lpxM</i> gene alters lipid A molecules by enhancing their modification via aminoarabinose	<i>lpxM</i> , <i>lpxR</i>	<i>E. coli</i> , <i>S. typhimurium</i> , and <i>K. pneumoniae</i>	Inactivation of <i>lpxM</i> triggers increased sensitivity to polymyxin due to the loss of L-Ara4N attachments. The increased binding ability of L-Ara4N to the lipid A helps to resist the polymyxin.	Maryland, Australia	[17, 39, 132]
The resistance of polymyxins via capsular polysaccharide	<i>Cpx</i> , <i>kpnEF</i>	<i>K. pneumoniae</i> , <i>Streptococcus pneumoniae</i> , and <i>P. aeruginosa</i>	Capsular polysaccharides protect organisms by inhibiting the interaction of Antimicrobial peptides with the cell surface. During capsular synthesis, the <i>kpnEF</i> plays a vital role via a multi-drug efflux pump <i>kpnEF</i> mutant that displayed a deficiency in capsular synthesis.	Spain, China	[47, 48, 62]
Efflux pump	Mutation in <i>acrAB</i> , <i>kpnEF</i> and <i>cpxAR</i>	<i>K. pneumoniae</i>	The AcrAB efflux pump is related to antimicrobial resistance. Mutation in <i>kpnEF</i> , a member of the Cpx regulon resulted in increased resistance to colistin. CpxR attachment with the efflux pump promoter subsequently helps to upregulate the expression of <i>cpxAR</i> mutant, and <i>kpnEF</i> .	Spain, India	[53, 55, 133]
Complete loss of LPS production	<i>lpxA</i> , <i>lpxC</i> and <i>lpxD</i>	<i>A. baumannii</i>	Alterations of amino acids within any of the three genes in the lipid A biosynthesis pathway: <i>lpxA</i> , <i>lpxC</i> , and <i>lpxD</i> resulted in the complete loss of lipopolysaccharide production, subsequently leading to colistin-resistant of <i>A. baumannii</i> .	Australia	[14]
The unique structural modification includes acylation, phosphorylation, and glycosylation of lipid A with hexosamine	<i>pEtN</i> and GalN	<i>A. baumannii</i>	<i>A. baumannii</i> having these genes <i>pEtN</i> and GalN modification that helps to resist colistin.	USA	[134]
LPS modification with the binding of 4-amino-L-arabinose to lipid A with a two-component system required for stimulation of the <i>amBCADTEF</i> operon modification	<i>parRS</i> , <i>colRS</i> , and <i>cprRS</i>	<i>P. aeruginosa</i>	In the presence of polymyxin, the <i>parR/parS</i> system can stimulate the <i>amBCADTEF</i> operon modification, that increased resistance to antibiotics. Mutation in <i>colR</i> or <i>colS</i> additionally obstructed the binding of 4-amino-L-arabinose to lipid A moiety, responsible for polymyxin resistance. Amino acids alteration in <i>colRS</i> and <i>cprRS</i> , regulates high-level polymyxin resistance in <i>P. aeruginosa</i> .	Canada, USA	[135, 136]

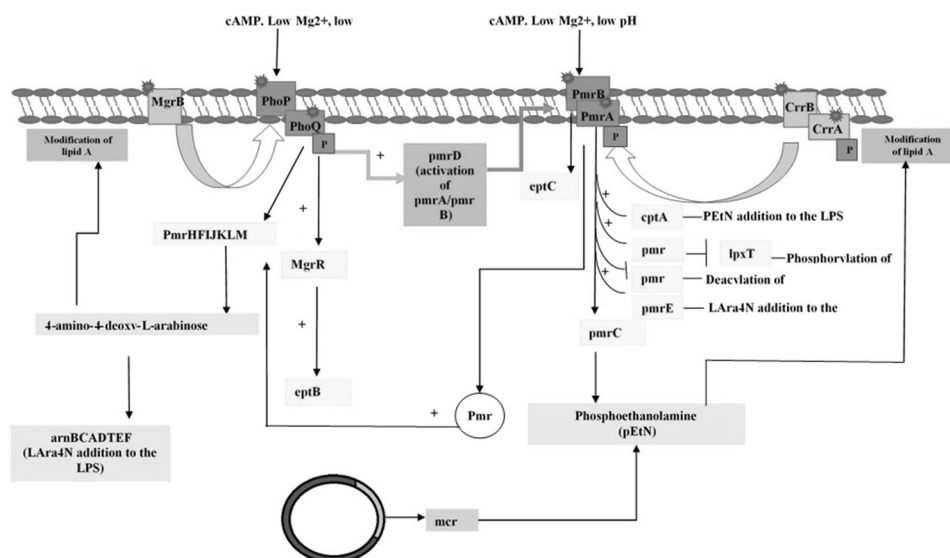


Fig. 1 Regulatory network modulation of lipid A moiety and plasmid-mediated pathways of colistin resistance in resistant bacteria. In presence of cationic peptide polymyxins, low Mg^{2+} , and Ca^{2+} , acidic pH, via the histidine kinase PhoQ and PmrB activates the two component systems (TCS) PhoP/PhoQ and PmrA/PmrB (the red star mark denotes phosphorylation), respectively. These TCS promotes the activation of the *arnBCADTEF* and *pmrCAB* operons helps to produce and addition of 4-amino-4-deoxy-L-arabinose (L-Ara4N) and phosphoethanolamine (PEtN) to lipid A, respectively. Furthermore, PhoP/PhoQ activates *pmrAB* through stimulation of *pmrD* gene, that activates

pmrA in-turn upregulates *arnBCADTEF* operon. MgrB with altered amino acid helps in overexpression of the *phoP-phoQ* operon, leading the upregulation of the *pmrHFJJKLM* operon, that aids in production of L-AraN. In addition, amino acids alteration in PmrB or in the response regulator PmrA, helps to activate PmrAB TCS. Successive upregulated *pmrCAB* operon promotes the addition of PEtN to lipid A. Plasmid mediated *mcr* gene encodes phosphoethanolamine transferase, responsible for resistance via the alteration in the phosphate groups of lipid A moiety in LPS

A moiety of the organisms leading to the acquisition of polymyxin resistance in *K. pneumoniae* has been described [49, 50].

Bacterial capsule formation depends on regulator molecules like Cpx and Rcs that regulate conjugative pilus expression and the synthesis of the capsule, respectively. The *ugd* gene is responsible for the synthesis of capsular polysaccharides and the L-Ara4N molecule. The active phosphorylated L-Ara4N is linked to polymyxin resistance [15, 44]. These regulatory molecules can also activate the efflux pump via KpnEF, which regulates the PhoPQ, two-component systems related to polymyxin resistance [44]. Loss of LPS in *Acinetobacter baumannii* may cause polymyxin resistance and induce sensitivity to other antibiotics [14, 51] (Table 1).

Resistance to polymyxin via overexpression of efflux pumps

Very few publications have suggested that efflux pump systems are also linked with polymyxin resistance, which can occur via activation of an efflux pump. Possibly, this is because of the target for the polymyxins is present outside the cell membrane. Some proteins, like KpnEF, AcrAB, and Sap (Table 1), have been reported as associated participant proteins in Enterobacterales for efflux pumps mechanisms

of polymyxin resistance [52, 53]. Mutation in the *kpnEF* gene was shown to induce polymyxin resistance. This gene belongs to Cpx regulation family associated with antimicrobial resistance [54]. Modification in the AcrAB–TolC complex that encodes AcrAB could result in polymyxin resistance. There have been reports that the expression of these proteins depends on the PhoPQ, the two-component systems [53]. Mutations in Sap proteins, a part of the *SapABCDF* operon, also aids in polymyxin resistance [55].

The role of intrinsic regulator RamA protein in resistance

RamA protein production is regulated by three genes named *ramA*, *romA*, and *ramR* [56]. Within this family, *ramR* gene acts as a repressor of the other two (*ramA* and *romA*) genes. Some Enterobacterales, like *K. pneumoniae*, possess a RamA regulator protein; this regulator molecule alters lipid A biosynthesis and is associated with permeability barriers. Modification in the *ramA* gene is linked to inducible resistance against polymyxins (Table 1). A report in 2015 revealed that increased levels of RamA could modify LPS on the bacterial surface, leading to induced polymyxin resistance [57]. Various genes are connected with lipid A biosynthesis pathway, including *lpxA*, *lpxC*, *lpxD*, *lpxB*, *lpxK*, *lpxL*, *lpxM*, and *lpxO* [58]. Study reports in *K. pneumoniae* showed that *ramA*

attaches directly to and stimulates the *lpxC*, *lpxO*, and *lpxL2* genes, leading to the modification of the lipid A moiety. Consequently, this modified lipid A moiety helps *Klebsiella* sp., gain the ability to survive polymyxin challenge [57]. The regulator *ramA* gene is also associated with the intact LPS permeability barrier and is thought to be linked to reduced susceptibility to polymyxin antibiotics.

Plasmid-mediated polymyxin resistance

Acquired resistance mechanisms to polymyxin are through chromosomal gene mutation as well as plasmid-mediated gene resistance associated with horizontal gene transfer [59] (Table 2). In 2015, the plasmid-mediated polymyxin resistance gene (e.g., mobile colistin resistance, *mcr-1*) was discovered from *E. coli* isolated from pigs in China [12] (Fig. 1). Shortly after this discovery, the *mcr-1* gene spread to various Gram-negative organisms, including *E. coli*, *K. pneumoniae*, *E. cloacae*, *E. aerogens*, *Citrobacter* sp., *Moraxella* sp., [10, 60] and has been reported globally from farm animal, food sectors, and aquatic environmental samples [10, 41, 61]. The *mcr-1* gene encodes a phosphoethanolamine lipid A transferase enzyme, which exhibits decreased susceptibility to polymyxin by catalyzing the attachment of pEtN to the lipid A moiety of LPS. This results in enhancement of cationic charge and reduction in the membrane interaction by polymyxin [22, 35]. The mechanism of *mcr-1* mediated resistance is the same as the intrinsic resistant mechanism of certain bacterial species as noted below. The polymyxin-resistant horizontal transferable *mcr-1* gene [60] (Table 2) has developed from *mcr-1* to *mcr-10* alleles [62]. All the allelic groups of *mcr* gene cause polymyxin resistance through PEtN transferases [63].

LptA and EptC (or *cptA*) the two PEtN transferases, identified from *Neisseria meningitidis*, *Serratia* sp., *Proteus* sp., *Burkholderia mallei*, and *Campylobacter jejuni*, respectively, resulting in innate resistance to polymyxins [22, 64]. The polymyxin-resistant *mcr-2* was recognized in *E. coli* from Belgium's cattle and porcine as well as in human vaginal swabs from China [64, 65]. The *mcr-3* gene was isolated in *E. coli* from swine in China [66]. The *mcr-4* gene was identified in *S. enterica* isolated from a pig in Italy [67, 68]. The *mcr-5* gene was detected in *S. enterica* serovar Paratyphi B from poultry in Germany [63]. The *mcr-5.3* gene variant was also identified in *E. coli* from an infected horse in South America [69, 70]. The *mcr-6* gene (also known as *mcr-2.2*) has been identified in *Moraxella* sp., isolated from pigs in Great Britain [71]. The *mcr-7* gene has been recognized in *K. pneumoniae* isolated from chickens in China [72]. The *mcr-8* gene was found in NDM-producing *K. pneumoniae* isolated from both pigs and humans in China [73]. In 2019, the novel *mcr* homolog, *mcr-9*, was identified from *S. enteritidis* serovar Typhimurium isolated in Washington, United States of America [74]. Finally, in 2020, *mcr-10*, a novel gene located on IncFIA plasmid was isolated from *Enterobacter roggenkampii* in China [75] (Table 2).

Methods for evaluating resistance to polymyxins

Colistin/ Polymyxin B susceptibility testing guides the appropriate management of multi-drug resistant bacterial infections. There are various phenotypic, susceptibility testing, and genotyping tools that are used to evaluate polymyxin resistance. Dilution methods include broth

Table 2 The list of colistin resistance plasmid-mediated *mcr* gene variants and their source

Gene	Year of Discovery	Allele/Genetic variants	Bacteria/isolated area	Sample origin	Length (bp)	References
<i>mcr-1</i>	1980, 2008	22	<i>E. coli</i> /China; <i>S. sonnei</i> /Vietnam	Animal, Human, Food	1626	[22, 68, 137]
<i>mcr-2</i>	2012, 2016	3	<i>E. coli</i> /Belgium; <i>E. coli</i> /China	Animal, Human	1617	[64, 65]
<i>mcr-3</i>	2005, 2016	30	<i>Aeromonas</i> /Germany <i>E. coli</i> /China;	Animal, Human	1626	[22, 65, 66, 138]
<i>mcr-4</i>	2008, 2013	6	<i>A. baumannii</i> /Brazil; <i>Salmonella enterica</i> Typhimurium/Italy; <i>E. coli</i> /Singapore	Human, Animal	1626	[67, 139, 140]
<i>mcr-5</i>	2011, 2018	4	<i>Salmonella enterica</i> serovar Paratyphi B/ Germany; NA/China	Animal, Human	1644	[63, 66, 69, 140]
<i>mcr-6</i>	2018	1	<i>M. pluranimalium</i> /UK	Animal	1617	[71, 141]
<i>mcr-7</i>	2015	1	<i>K. pneumoniae</i> /China	Animal	1620	[72]
<i>mcr-8</i>	2015, 2018	4	<i>K. pneumoniae</i> /China; <i>K. pneumoniae</i> / Algeria	Animal, Human	–	[73, 142]
<i>mcr-9</i>	2010, 2016	2	<i>S. enterica</i> serotype Typhimurium/USA; <i>K. oxytoca</i> /Sweden	Human, Animal	–	[74]
<i>mcr-10</i>	2020	2	<i>E. roggenkampii</i> /China	Human	1620	[75]

dilution, broth microdilution, broth macrodilution, agar dilution method, disk-diffusion test, E-test, detection of enzymes responsible for polymyxin resistance, and molecular practices include the use of PCR and whole-genome sequencing to screen for genes that confer resistance to the polymyxins [76, 77]. The selection of resistant bacteria through its modified outer membrane properties by capillary electrophoresis provides the benefit of qualitative detection of polymyxins resistant bacteria [78], micromax assay that detects the release of nucleotides by the cell damage caused by polymyxin [18], and the use of Polymyxin NP (Nordmann/Poirel) test that detects the growth of the bacteria in the presence of defined polymyxins (both polymyxin B and colistin) dosage [78]. Screening for polymyxin-resistant isolates requires culture media containing polymyxin in lower concentrations, like $2\text{--}4\text{ }\mu\text{g ml}^{-1}$. Agar culture media to detect polymyxin-resistant isolates are not available commercially [79]. In-house developed media such as LB non-selective agar, MRS agar, MacConkey medium with polymyxin 2 or $4\text{ }\mu\text{g ml}^{-1}$ have been developed to isolate polymyxin-resistant strains [80, 81]. These media can be used directly for culturing samples to detect the growth of polymyxin-resistant strains [70, 82]. Broth medium containing polymyxin can also be effective for the detection of resistant isolates. Disk susceptibility testing to determine polymyxin resistance is unacceptable because the diffusion of polymyxins through the agar is very low. In both gradient and well diffusion tests, polymyxins tend to bind to the plastic, such as microtiter plates, tips which lead to variable measurements. The broth microdilution method can be used for the determination of polymyxin-resistant strains [83, 84]. Based on the polymyxin breakpoints formed by the international standard-setting organization such as EUCAST (European Committee on Antimicrobial Susceptibility Testing) and CLSI (Clinical and Laboratory Standards Institute), test isolate can be reported as susceptible, resistant, and intermediate isolate [76, 77]. According to the EUCAST guidelines for polymyxin susceptibility, the MIC breakpoint is $\leq 2\text{ }\mu\text{g ml}^{-1}$, and polymyxin resistance is $>2\text{ }\mu\text{g ml}^{-1}$ [77]. The broth microdilution method is time-consuming and alternate methods are used for easy detection. Polymyxin enzyme strips infused with increasing antibiotic concentrations can be used to determine the levels of resistance. This method could guide the treatment decisions for critical cases. [79]. The thin enzyme strip is an inert, porous strip coated with polymyxin. The MIC reading is printed on both the sides of the enzyme strip that ranges the value from $0.06\text{ to }64\text{ }\mu\text{g ml}^{-1}$ and $0.016\text{ to }256\text{ }\mu\text{g ml}^{-1}$ [18, 77]. The enzyme strip placed on the medium inoculated with the test organism will help identify MIC of the resistant isolate as the polymyxin immobilized in the strip diffuses into the medium ranging across 15-fold dilution of a conventional MIC method [79].

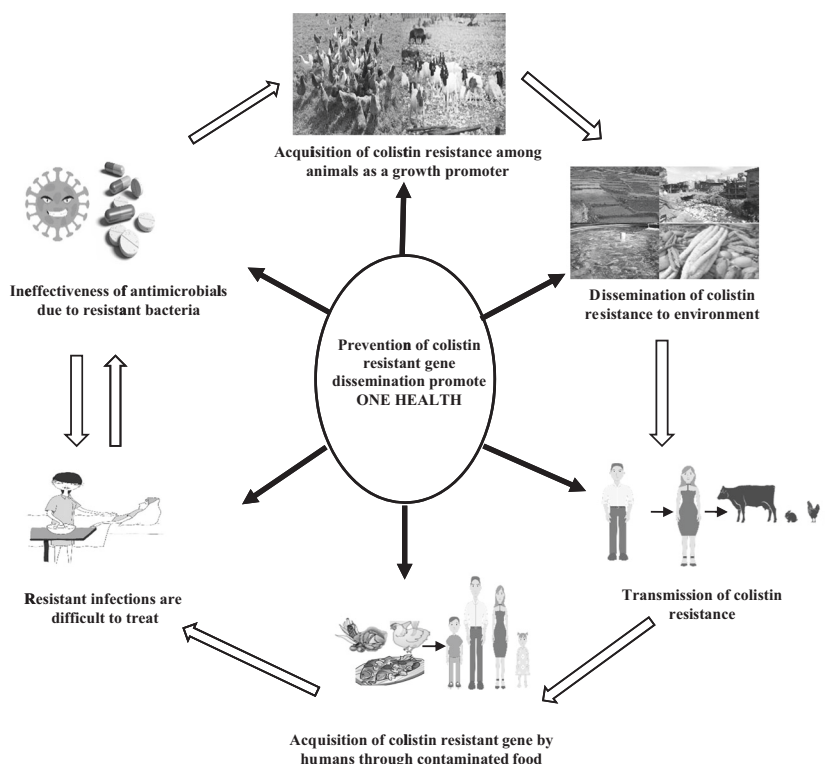
Molecular screening and genotyping can detect the presence of polymyxin resistance genes. The occurrence of *mcr-1* and *mcr-2* genes precisely corresponds to resistance to polymyxin. The detection of *mcr-1* gene in the resistant isolates using SYBR green-based RT-PCR is a sensitive, specific, and rapid molecular method [85]. These genotypic tests are complementary to phenotypic assay and can confirm the presence of the resistance genes in isolates. Whole-genome sequencing is one of the genotypic methods to evaluate the occurrence of colistin/ polymyxin B resistance genes but requires a reference database [86].

Polymyxin resistance in food bacteria and the related food chain

Since the 1960s, the oral usage of polymyxins have been approved for use in veterinary medicine and in farm animals for growth promotion *via* changing the intestinal microflora, and also to prevent and treat infections caused by Enterobacterales [20, 87]. During the past decades an assortment of bacterial resistances due to selective antibiotic pressure and the dissemination of antimicrobial resistance through bacteria from various niches has been observed (Fig. 2). The selective pressure created by the massive and inappropriate usage of polymyxins have resulted in higher polymyxin resistance worldwide among the bacterial isolates of food-producing animals [88]. Approximately twenty-six European countries reported a significant positive relationship between the use of polymyxin in food-producing animals as a growth promoter and polymyxin resistance in *E. coli* [89–91]. *E. coli* harboring polymyxin-resistant gene was discovered in Bangladesh from various samples such as street food, water, hand rinse of street food sellers, and from the gut of healthy individual's [92]. Some recent studies suggest that the prevalence of polymyxin-resistant bacteria isolated from poultry, pigs, cattle, food products originated from livestock as well as from environmental sources and humans [93–97].

Enterobacterales having polymyxin-resistant gene are prevalent in both animals and humans and are also isolated from vegetables, rivers, pet animals, wildlife, and livestock [63, 86, 93, 97, 98]. These studies report that food producing animals are the principal reservoir of *mcr-1* and *mcr-2* genes [12, 18, 64]. Those findings describe that an ecosystem without any barriers could be a possible route for spreading polymyxin-resistant bacteria containing the *mcr-1* gene *via* the food chain to humans (Fig. 2). *Actinobacillus pleuropneumoniae* isolated from the pig was involved in mobilizing the currently identified *mcr-1* gene [22, 99]. Therefore, the unnecessary use of polymyxin in food producing sectors must be reduced to prevent the transmission of resistance genes through the food chain to humans [21, 100].

Fig. 2 Colistin resistant bacteria/ gene transmission through food chain: The use of antibiotics as a growth promoter can induce acquisition of antibiotic resistance among animals. This can directly spread antibiotic resistance into the surrounding environment. The exposure of humans with animals and contaminated food endorses the acquisition of colistin resistance. The resistant infections caused will be ineffective for the treatment



To understand the origin and transmission of *mcr* genes, one must consider a ‘One Health’ strategy to control polymyxin usage to prevent further dissemination. Such an approach needs acceptance worldwide to institute interdisciplinary communication among concerned experts in all the sectors (Fig. 2). The spread of antimicrobial-resistant genes cannot be controlled without understanding and stopping the actual problem of indiscriminate usage of polymyxins in animal farming. A research study advised that the ‘One Health’ strategy must be used to understand antimicrobial resistance better, as a global public health threat, to delay/stop antimicrobial resistance, and safeguard human life [101]. European Food Safety Authority (EFSA) applied the “One Health Approach”, which describes the strategic framework for combating antimicrobial resistance, to prohibit the inappropriate use of polymyxins in animal farming [102–104]. The prevalence of polymyxin-resistant genes in bacteria isolated from animal origin remains the cause of concern. The proportion of polymyxin-resistant serotypes in healthy animals contributes to the rate of resistant gene dissemination. Enteropathogenic *E. coli* is considered as the leading indicator organism carrying colistin/ polymyxin B resistance and beta-lactamase genes in animal intestinal infections [72, 105]. Polymyxin-resistant *E. coli*, from raw meat, also harbored was also found to coexist with other resistance genes responsible for tetracycline, quinolone, trimethoprim, and beta-lactams resistance [106]. Further studies showed that various organisms like *Aeromonas veronii* and *E. coli* isolated from

retail chicken meat, swine farms, and chicken feces were found to contain polymyxin resistance genes [107–110]. A report from Tunisia demonstrated extended-spectrum beta-lactamase-producing *E. coli* isolated from caprine and bovine milk samples and feces of bovine together with genes encoding *mcr*-1 to 5 responsible for polymyxin resistance [110]. Significant risk was found in disseminating polymyxin resistance through conjugative plasmid containing *mcr*-1 gene in Enterobacterales isolated from intestinal microbiota of poultry [111, 112]. Indiscriminate use of antibiotics has been found in different sectors, including retail meat, poultry, and aquaculture, which plays a significant role in disseminating global antibiotic resistance [113, 114]. One health strategy can be implemented in mitigating the dissemination of polymyxin resistance in the food chain.

The usage of polymyxin in India

In India, antibiotics are used for nontherapeutic purposes, especially as a growth promoter, in apiary, poultry, aquaculture, and agriculture to promote the harvest [115, 116]. The polymyxin (colistin sulfate) used as a growth promoter is commercially available in India and can be obtained without any prescription [117]. Reports from India also identified *Aeromonas* sp., and *Salmonella* sp., from different samples like water, fish, diseased catfish, retail markets, and poultry farms that are resistant to different antibiotics,

including polymyxin [21, 118]. The indiscriminate use of such antibiotics in various poultry farms of India has helped several pathogens to develop an extensive resistance towards antibiotics [119]. It has been reported that five animal pharmaceutical companies promote the production of antibiotics like polymyxins for supporting growth and for metaphylactic purposes. As per the investigation held by the Bureau of Investigative Journalism of London, it was pointed out that chickens raised in India are treated with potent antibiotics. Venky's, the chief supplier of chicken products in Andhra Pradesh, India, has stated that the common practices of using polymyxin antibiotics are for therapeutic purposes [120]. This data indicates that there is a misuse of antibiotics in the field of veterinary and animal husbandry as medication and growth promoters respectively. Gaining antibiotic resistance by organisms is the consequence of selective antibiotic pressure; resulting from indiscriminate use of polymyxins and other antibiotics.

Risk factors for polymyxin resistance

It is essential to know the risk factors associated with antimicrobial (polymyxin) resistance, as polymyxins are used as a last resort drug for the therapy of multidrug-resistant infections [99, 121]. It is on the WHO list of essential medicines. Polymyxin resistance is an independent risk factor enhancing death from inadequate treatment due to the presence of polymyxin-resistant infections [122, 123]. The risk factors of polymyxin resistance have not been well studied. However, retrospective studies using logistic regression models have been employed to produce univariate and multivariate forecasters of polymyxin resistance, which showed that there is a significant relationship between the isolation of polymyxin-resistant organisms with other factors, such as prolonged hospitalization and increased comorbidity, and antibiotic exposure [122, 123]. For example, one case study showed that after lower urinary tract surgery, multiple episodes of hospitalization, stay at the hospital for more than two weeks, exposure to more than three classes of antimicrobials were significant risk factors linked with polymyxin-resistance among uropathogenic isolates of *E. coli* and *K. pneumoniae* [124, 125]. In terms of accessing the potential risk of polymyxin resistance, the contribution and their interactions of polymyxin resistance in animals, food, the environment, and humans have to be studied.

Conclusion

Polymyxins are needed to treat seriously ill individuals who have multidrug-resistant bacterial infections. Therefore,

there is a need to prevent the transmission of polymyxin-resistance in the food chain, and inappropriate antibiotic usage in animal farming should be stopped. To prevent further dissemination of polymyxin resistance, it is essential to monitor, screen, and analyze the spread of polymyxin resistance from animals to humans and humans to humans to combat and inhibit antibiotic-resistant bacterial infections. Additionally, it is required to observe the epidemiology of disseminating polymyxin resistance among Gram-negative bacteria and to promote studies that aid in the proper use of polymyxin in all sectors. The one health approach needs the mutual involvement of numerous disciplines and healthy cooperation between medical doctors, veterinarians, food producers, and other scientific and environmental health professionals.

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Compliance with ethical standards

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